

Exercises 5

Exercise 1 (Quantifier Elimination in PA). Apply quantifier elimination as seen in the Lectures to the following formulas:

1. $\exists x, y. 2x + 3y < 7 \wedge x < y$
2. $\exists x, y. 2x + 3y < 7 \wedge y < x$
3. $\exists x, y. 3x + 3y < 8 \wedge 8 < 3x + 2y$
4. $\exists x, y. x = 2y \wedge \exists z. x = 3z$

Exercise 2 (Satisfiability algorithm for Presburger arithmetic). Consider the formula $F(x)$ given by

$$F(x) = \bigwedge_{i=1} a_i < x \wedge \bigwedge_{j=1} x < b_j \wedge \bigwedge_{i=1} K_i | (x + t_i).$$

Note that the terms a_i, b_j, t_i may in general contain variables other than x .

1. First, assuming that all a_i, b_j, t_i are integer constants, give an algorithm that, given any formula of the form above, returns:
 - a value for x such that $F(x)$ holds, if one exists, and
 - “UNSAT” if no such value exists
2. In the case that a_i, b_j, t_i may contain other variables, give a recursive algorithm that returns:
 - an assignment to each variable (including x) such that $F(x)$ holds, if one exists
 - “UNSAT” if no such assignment exists.

Exercise 3 (Quantifier elimination for rationals). In this exercise we will devise a quantifier elimination method for rational numbers. We consider formulas over the signature $(\mathbb{Q}, <, \leq, =, +, -)$, i.e. with constant symbols among $\mathbb{Q}$, interpreted over the standard structure of rational numbers.

1. Show that for any formula F , there exists a formula F_1 such that

$$F \iff Q_1 x_1, \dots, Q_n x_n, Q_{n+1} y. F_1$$

where each Q_i is either $\exists$ or $\neg\exists$, i.e. existential quantifiers that can be separated by negations and where F_1 is built only from $(\wedge, \vee, \mathbb{Q}, <, =, +, -, k \cdot _)$. In particular, it is quantifier-free and contains no negations.

2. Do we need to add a divisibility relation as in the Presburger arithmetic case? Why, or why not?
3. Show that there exist a formula F_2 such that $F_1 \iff F_2$ and every atom of F_2 containing y is of the form:

$$t < y \quad \text{or} \quad y < t \quad \text{or} \quad t = y$$

for some term t .

4. Show that there exists a quantifier-free formula F_3 such that

$$(\exists y.F_2) \iff F_3$$

Exercise 4 (Presburger Arithmetic without divisibility). Following the steps below, show that Presburger Arithmetic without the divisibility predicate does *not* admit quantifier elimination.

1. For a formula $F(y)$ with one free variable y , define the set S_F as

$$S_F = \{n \in \mathbb{N} \mid F(n)\}$$

i.e. the set of positive integers that satisfy F .

Show that there exists a formula $F(y)$ such that S_F is neither finite nor co-finite in $\mathbb{N}$ (i.e. S_F and $\mathbb{N} \setminus S_F$ are both infinite).

2. Show that for any *quantifier-free* formula $G(y)$, S_G is either finite or co-finite in $\mathbb{N}$.
3. Using the results from the previous parts, show that Presburger Arithmetic without the divisibility predicate does not admit quantifier elimination.

Exercise 5 (Structure of sets). Consider the structure $(\mathcal{P}(\mathbb{N}), \subseteq, =, \cap, \cup, \cdot^c)$ whose base set is the set of all sets of natural numbers and where $\cdot^c$ denotes set complementation with respect to the universe.

Is it possible to eliminate quantifiers from arbitrary first-order formulas on this structure? For example, $\exists B. A \subseteq B \wedge B \subseteq C$ is equivalent to $A \subseteq C$.

Either:

1. Show that this structure admits quantifier elimination by giving a procedure to compute an equivalent quantifier-free formula from any given formula, **or**
2. Produce a quantified first-order logic formula that has no equivalent formula without quantifiers, and prove this property.

Exercise 6 (A rational arithmetic formula). Consider the following formula $G(x, z)$ where the variables range over rational numbers $\mathbb{Q}$:

$$\forall y.((x < y \wedge y < z) \longrightarrow \forall u.(x \neq u + u + u))$$

Find a quantifier-free formula equivalent to $G(x, z)$.